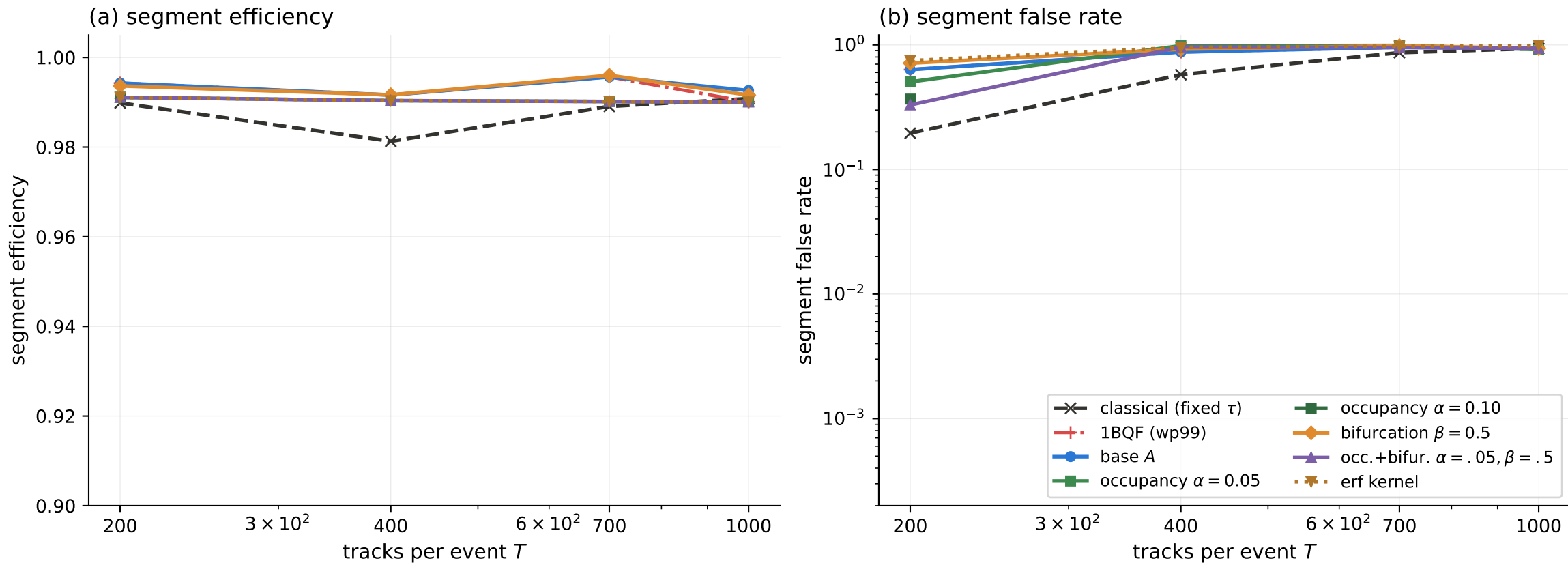


Segment metrics against track density — heavy noise (fitted response, $d = 40$; quantum at wp99, classical at fixed τ)



$\sigma_{\text{scatt}} = 10^{-4}$, $\sigma_{\text{res}} = 20 \mu\text{m}$, drop 1% · formula $\varepsilon = 6.31 \text{ mrad} \cdot \gamma = 3$, $\delta = 1$, step kernel · occupancy $\alpha = 0.05$ · fork $\beta = 0.5$, $\varepsilon_B = \varepsilon \cdot \text{fitted} = \text{moment-space ridge}$, μ by eff@far $\leq 1\%$ · T reps: 10 (200) / 5 (400) / 3 (700, 1000) · 1BQF = matched cosine $|\cos(\pi\lambda/2s')|$ on the same events